

Jouri Aldaghma

Computer Science (BSc) | Artificial Intelligence Concentration | Effat University

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EDUCATION

Effat University | Aug 2022 - Expected Jun 2026

Bachelor of Science in Computer Science | AI Concentration | Jeddah, Saudi Arabia

- GPA: 3.96 / 4.00

Edugate International School | Graduated May 2022

American High School Diploma | Jeddah, Saudi Arabia | GPA: 4.0 / 4.0

HONORS & AWARDS

First Place — Effat Undergraduate Research Forum (5th Edition) | May 2024

- Awarded for 'Analyzing Security Threats and Vulnerabilities in IoT Systems within Smart Cities.'

Dean's List — Effat University | 2022, 2023, 2024

- Recognized for outstanding academic achievement for three consecutive years.

Academic Merit Scholarship — Effat University | 2022 - Present

- Awarded 50% tuition scholarship based on academic excellence, maintained until graduation.

SELECTED PAPERS

4 of 8 published articles listed below. Full list: researchgate.net/profile/Jouri-Aldaghma

A Multi-Layered Adaptive Framework for Adversarially Robust AI in Cybersecurity Applications

Dec 2025

- Analyzes adversarial ML threats (2022-2025) and proposes a Multi-Layered Adaptive Defense Framework combining detection, incremental retraining, and explainable auditing for cybersecurity AI.

Advancing Cybersecurity with Digital Twin Technology

Apr 2025

- Systematic analysis of Digital Twin applications in cybersecurity; demonstrated 97.5% detection accuracy and 1.5s latency in simulated IoT-AI environments.

Advancements in Kernel Concurrency: Leveraging Machine Learning for OS Innovation

Sep 2024

- Innovative ML approaches for kernel concurrency: test prioritization, bug detection, and adaptive synchronization in operating systems.

Analyzing Security Threats and Vulnerabilities in IoT Systems within Smart Cities [1ST PLACE - UGRF #5]

May 2024

- Comprehensive threat modeling of IoT attack surfaces in smart-city infrastructure. Awarded First Place at the 5th Effat Undergraduate Research Forum.

PROJECTS

A3L: Adaptive Adversarial Active Learning for CTI Classification

Dec 2025

Technologies: Python | DistilBERT | SecBERT | SecRoBERTa | HuggingFace | FGSM | PyTorch

- Designed a multi-agent framework (Detection, Selection, Retraining, Audit agents) for adversarially robust CTI classification using MITRE ATT&CK; labels on the AnnoCTR corpus.

- Implemented Adversarial Sample Utility scoring and epsilon curriculum [0.02-0.10] to minimize labeling cost while maximizing robustness; benchmarked against 8 defense strategies including Core-Set, Entropy, and Full Adversarial Training.

SecureTwin — Cybersecurity Digital Twin Platform

Dec 2025

Technologies: System Architecture | Network Simulation | Threat Modeling

- Designed a comprehensive cybersecurity testing platform architecture using digital twin methodology.

- Created detailed system diagrams and performance simulations for 10,000-node environments.

IoT Cyber-Attack Detection Using Machine Learning

May 2025

Technologies: Python | Random Forest | XGBoost | PCA | SMOTE | Scikit-learn

- Built end-to-end ML pipeline for detecting cyber-attacks in IoT network traffic.

- Applied feature engineering, SMOTE balancing, and PCA — achieved ~98% detection accuracy.

Educational Institution Network Design

Aug 2024

Technologies: Cisco Packet Tracer | VLAN | Firewall | Routing

- Designed and implemented a network architecture for a multi-campus educational institution, supporting secure communication between devices.

TECHNICAL SKILLS

AI / ML: Python, PyTorch, TensorFlow, Scikit-learn, HuggingFace Transformers, NLP, Adversarial ML

Data Science: Pandas, NumPy, Matplotlib, Seaborn, Jupyter Notebook

Cybersecurity: Adversarial Robustness, Digital Twin Security, IoT Security, Threat Modeling, Intrusion Detection

Systems & Networks: C++, Cisco Packet Tracer, Linux, Hadoop, HDFS, MapReduce

Web & Tools: HTML, CSS, JavaScript, PHP, Git, Google Colab, Power BI

Languages: Arabic (Native), English (Professional)